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EXPERIEMNT 25

Prolog Program: Monkey-Banana Problem

Aim

To write a **Prolog program to implement the Monkey-Banana problem**, where a monkey tries to get a banana hanging from the ceiling by using a box.

Algorithm

1. Represent the positions of the monkey, box, and banana.
2. The monkey can **walk** from one position to another.
3. The monkey can **push the box** to another position.
4. The monkey can **climb onto the box**.
5. When the monkey is on the box and at the banana position, it can **grasp the banana**.
6. Display the sequence of actions required to get the banana.

Prolog Code

% Monkey Banana Problem

% Initial state

initial_state(state(door, window, floor, no)).

% Monkey walks from door to window

move(state(door, window, floor, no),

```
state(window, window, floor, no),  
walk).
```

% Monkey pushes box from window to middle

```
move(state(window, window, floor, no),  
state(middle, middle, floor, no),  
push).
```

% Monkey climbs onto the box

```
move(state(middle, middle, floor, no),  
state(middle, middle, onbox, no),  
climb).
```

% Monkey gets the banana

```
move(state(middle, middle, onbox, no),  
state(middle, middle, onbox, yes),  
get_banana).
```

% Solve the problem

solve(Actions) :-

```
initial_state(S0),  
move(S0, S1, A1),  
move(S1, S2, A2),  
move(S2, S3, A3),
```

move(S3, S4, A4),

S4 = state(middle, middle, onbox, yes),

Actions = [A1, A2, A3, A4].

Input

solve(X).

Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/Exppp 25.pl').  
compiling C:/Users/MOSHIYA/Downloads/Exppp 25.pl for byte code...  
C:/Users/MOSHIYA/Downloads/Exppp 25.pl compiled, 33 lines read - 3050 bytes written, 16 ms  
  
yes  
| ?- solve(X).  
  
X = [walk,push,climb,get_banana] ?
```

Result

Thus, the Monkey-Banana problem was successfully implemented in Prolog, and the sequence of actions performed by the monkey to obtain the banana was displayed.